Modulus Functions: Inequalities

Questions 1–12

Question 1.

a) On the same set of axes (you do not need to label the points of intersection) sketch $y = |x^2 - 3x|$ and $y = \frac{5}{2} - \frac{1}{2}x$.

b) Hence solve $|x^2 - 3x| > \frac{5}{2} - \frac{1}{2}x$ giving your answer in interval notation.

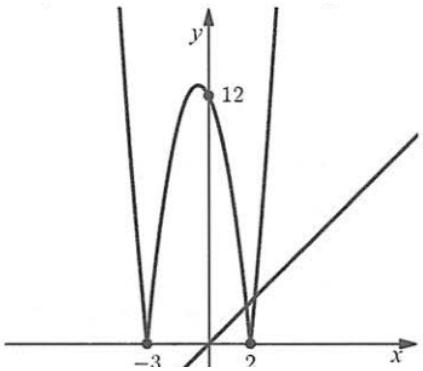
Question 2. Solve each of the following, giving your answer in set notation:

a) $|x^2 - 4| > 3$

b) $|x^2 - 2x - 3| \leq 5$

c) $|x^2 - x - 6| > 2x$

Question 3. The following graph shows the curves $y = |ax^2 + bx + c|$ and $y = x$, where a, b and c are non-zero integers.



a) Find the possible values of a, b and c.
b) Using your answers for part (a), determine the values for which $|ax^2 + bx + c| < x$ giving your answer in set notation.

Question 4.

a) On the same set of axes (you do not need to label the points of intersection) sketch $y = |x^3 + x^2 - 4x - 4|$ and $y = 10x + 20$.
b) Hence solve $|x^3 + x^2 - 4x - 4| > 10x + 20$ giving your answer in interval notation.

Question 5. Solve each of the following, giving your answer in set notation:

a) $|x^3 - x| > 3x$

b) $|x^3 - 2x| > x$

c) $|x^3 - 4x| \leq 3$

Question 6.

a) On the same set of axes (you do not need to label the points of intersection) sketch $y = \left|\frac{2x+1}{x-1}\right|$ and $y = x + 3$.

b) Hence solve $\left|\frac{2x+1}{x-1}\right| > x + 3$ giving your answer in set notation.

Question 7. Solve each of the following, giving your answer in set notation:

a) $\left|\frac{3x-2}{x+1}\right| > 2$

b) $\left|\frac{2x-1}{x-2}\right| < 3x+2$

c) $\left|\frac{x^2-4}{x-1}\right| \leq 4$

Question 8. Solve each of the following, giving your answer in interval notation:

a) $1 - |\cosh x| < 0$

b) $|1 - \sinh x| \leq 1$

c) $|\sinh x| + \cosh x \geq 2$

Question 9. Given that $f(x) = |-1 + \ln x|$ for $x > 0$, find the values for which $f(x) \leq 2$. Give your answer in set notation.

Question 10. Given that $f(x) = \left|\frac{1}{|x|}\right|$, find the values for which $f(x) \geq 1$. Give your answer in set notation.

Question 11. Given that $f(x) = \left|\cos x - \frac{1}{2}\right|$ for $-\pi < x \leq \pi$, find the values for which $f(x) \geq \frac{1}{2}$. Give your answer in interval notation.

Question 12. Given that $f(x) = \left|\frac{ax-1}{x-b}\right|$ for positive constants $1 < a < 4$ and $b > 1$, find the values for which $f(x) \geq 4$. Give your answer in set notation in terms of a and b.